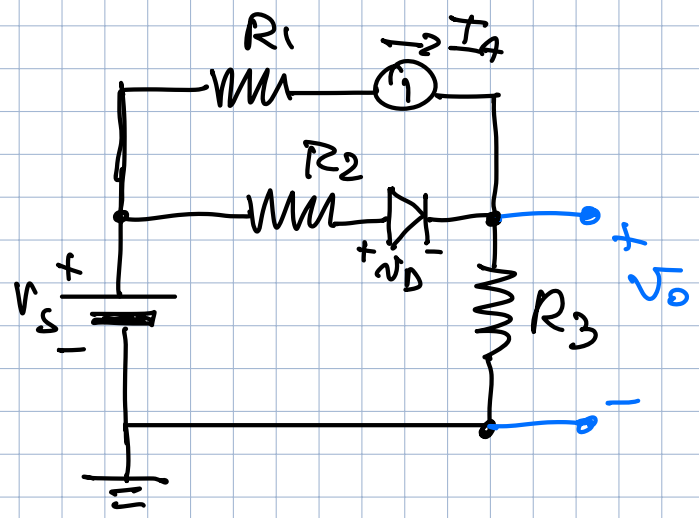


$V_{ON} = 1V$

$R_1 = 5\,K\Omega$

$R_2 = 2\,K\Omega$

$R_3 = 6\,K\Omega$

$I_A = 2\,mA$

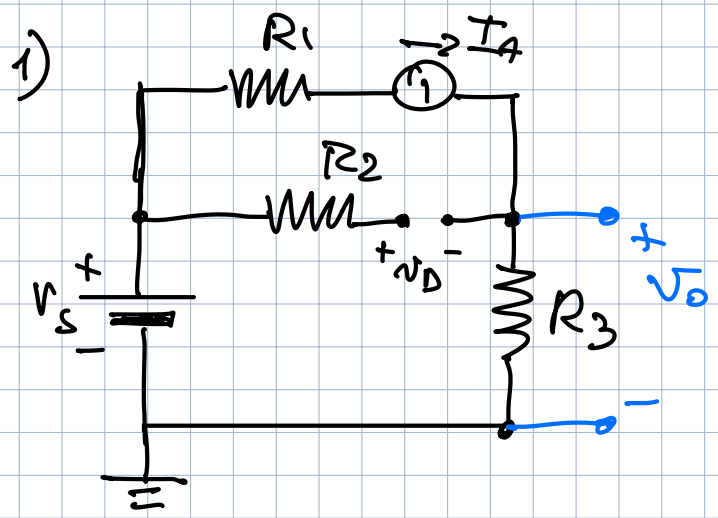
TROVARE

1) $v_o = f(v_s)$

2) CON $V_s = 2V$

trovare $v_o = f(I_A)$

1)



HP DIODO "OFF"

$v_o = I_A \cdot R_3 = 12V$ (COSTANTE)

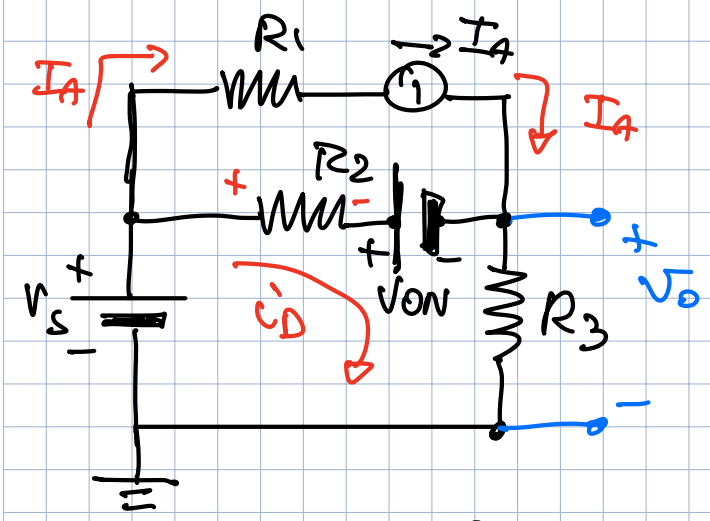
$v_D = V_s - v_o < V_{ON}$

CONDIZIONE X AVERE D = OFF

$\Rightarrow$ $V_s < 13V$ $\Rightarrow$ DIODO OFF

$v_o = 12V$ $\quad V_{sp} = 13V$

CON $V_s > 13V$ $\Rightarrow$ DIODO "ON"



$-V_s + R_2 i_D + V_{ON} + R_3(i_D + I_A) = 0$

$i_D(R_2 + R_3) = V_s - V_{ON} - R_3 I_A$

$i_D = \dfrac{V_s}{R_2 + R_3} - \dfrac{V_{ON} + R_3 I_A}{R_2 + R_3} > 0$

$V_s - (V_{ON} + R_3 I_A) > 0$

$V_s > 1V + 12V = 13V$ OK

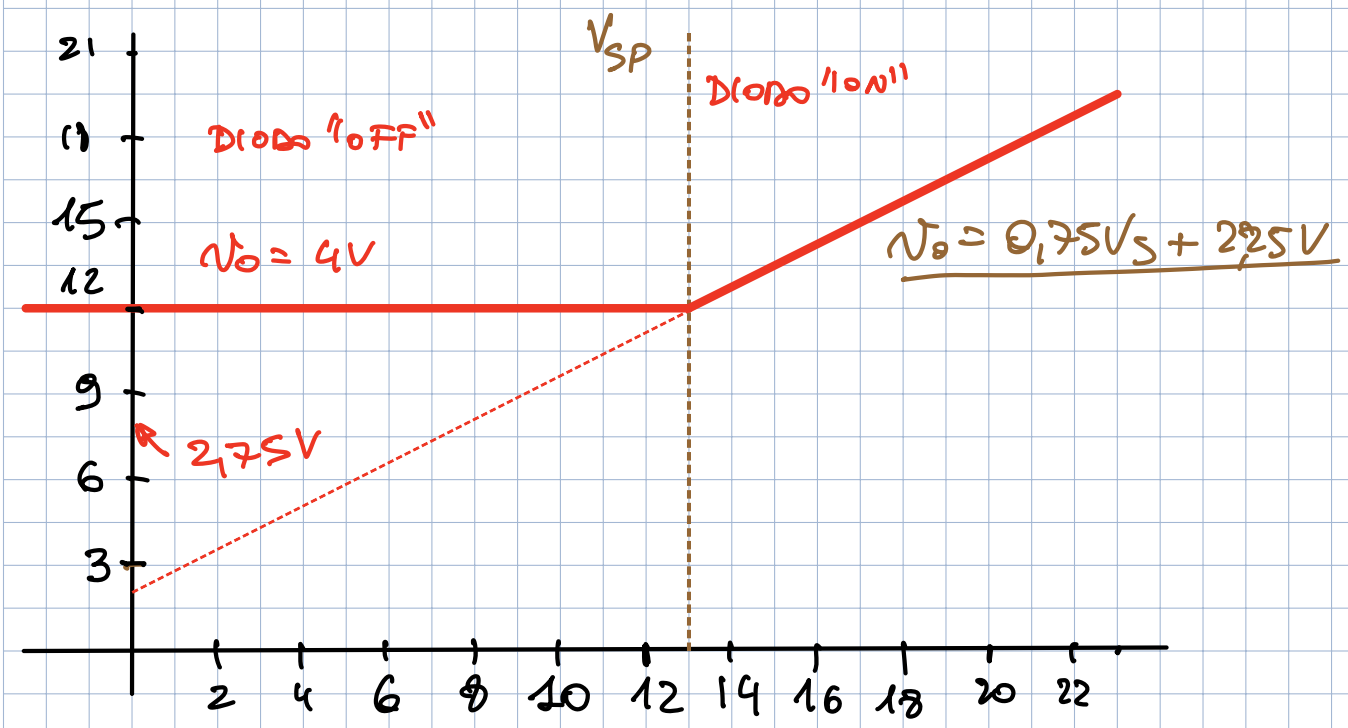
HO VERIF.

$$v_o = R_3(i_D + I_A) = V_s \frac{R_3}{R_2 + R_3} - \frac{(V_{ON} + R_3 I_A)}{R_2 + R_3} R_3 + R_3 I_A$$

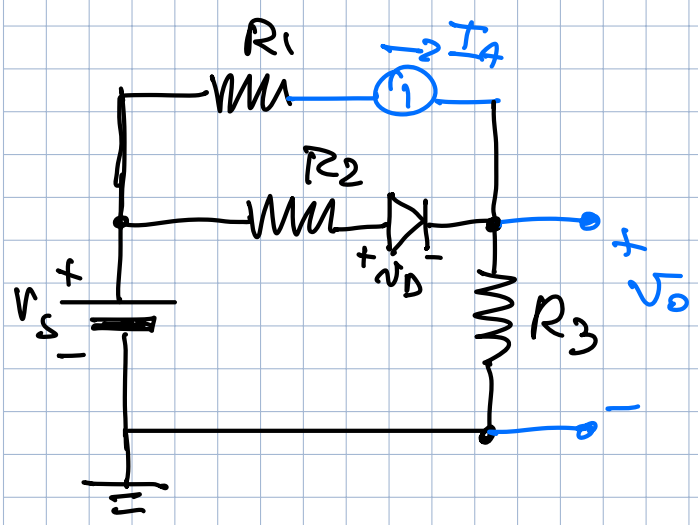
$$v_O = V_S \frac{6}{8} - 13V \cdot \frac{6}{8} + 12V$$

$$= \frac{3}{4} V_S - \frac{39}{4} + 12 = 0{,}75 V_S + 2{,}25$$

$v_O(13V) = 12V$ OK RACCORDO



2)



$V_{ON} = 1V$

$R_1 = 5 K\Omega$

$R_2 = 2 K\Omega$

$R_3 = 6 K\Omega$

$V_S = 13 V$

$v_O = f(I_A)$?

HP DIODO OFF

$v_D = V_S - R_3 I_A < V_{ON} \Rightarrow I_A > \frac{V_S - V_{ON}}{R_3} = 2{,}0 \text{ mA}$

$v_O = I_A R_3$

$I_{SP} = 2 \text{ mA}$

$$v_O(I_{SP}) = v_O(2\,mA) = 12\,V$$

CON $I_A < I_{SP}$ $\Rightarrow$ DIODO "ON"

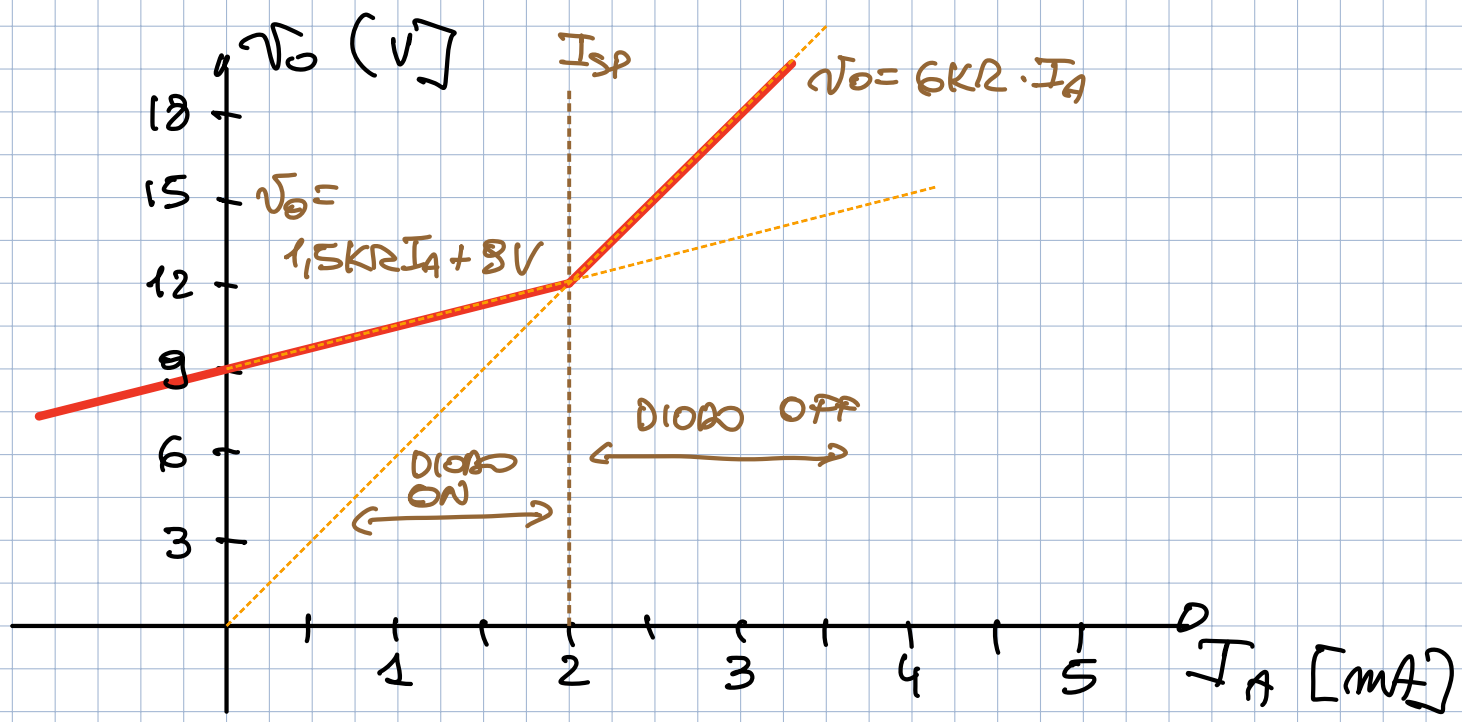
DA "SOPRA":

$$v_O = R_3(i_D + I_A) = V_S \frac{R_3}{R_2+R_3} - \frac{(V_{ON} + R_3 I_A)R_3}{R_2+R_3} + R_3 I_A$$

$$v_O = (V_S - V_{ON})\frac{R_3}{R_2+R_3} + R_3 I_A\left(1 - \frac{R_3}{R_2+R_3}\right)$$

$$= 12 \cdot \frac{6}{8} + I_A \frac{R_3 \cdot R_2}{R_2+R_3}$$

$$= 1{,}5\,k\Omega\, I_A + 9\,V$$

$$v_O(I_{SP}) = 1{,}5\,k\Omega \cdot 2\,mA + 9\,V = 12\,V$$

OK RACCORDO